T3 A_-4: full general UV Lagrangian and RGEs

Matchete/RGBeta symbolic output in mathematical notation

A_-4

Additional quartics: 0, additional bilinears: 0, extra Yukawa: none.
General matching and Wilson coefficients are not computed in this symbolic report.

Coupling and interaction notation

symbol	interaction / definition
g_1	$D_\mu \supset i g_1 Y B_\mu$
g_2	$D_\mu \supset i g_2 T^A W_\mu^A$
g_3	$D_\mu \supset i g_3 t^A G_\mu^A$
Y_u	$-\bar{Q} Y_u \widetilde{H} u + \text{h.c.}$
Y_d	$-\bar{Q} Y_d H d + \text{h.c.}$
Y_e	$-\bar{L} Y_e H e + \text{h.c.}$
y_1	$L F S_1 + \text{h.c.}$
y_2	$L F S_2 + \text{h.c.}$
M_F	$-\bar{F} M_F F$
T	$\text{Tr}(Y_e^\dagger Y_e + 3 Y_u^\dagger Y_u + 3 Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\text{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\text{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2	$m_1^2 S_1^\dagger S_1$
m_2^2	$m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2}\lambda_1 (H^\dagger H)^2$
$\lambda_{S_1}^{(1)}$	$\frac{1}{2}\lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$
$\lambda_{S_2}^{(1)}$	$\frac{1}{2}\lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$
$\lambda_{HS_1}^{(1)}$	$\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$
$\lambda_{HS_2}^{(1)}$	$\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$
$\lambda_{HS_1}^{(A)}$	$\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$
$\lambda_{HS_2}^{(A)}$	$\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$
λ_5	$\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$
$\lambda_{12}^{(1)}$	$\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	$\lambda_{12}^{(\times)} (S_1^\dagger S_2)(S_2^\dagger S_1)$
$\lambda_{S_1}^{(A)}$	$\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1)(S_1^\dagger T^A S_1)$
$\lambda_{S_2}^{(A)}$	$\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2)(S_2^\dagger T^A S_2)$
α	$Y_{S_1} = \alpha/2,\ Y_F = (\alpha+1)/2,\ Y_{S_2} = (\alpha+2)/2$
$C_2(d)$	$(d^2-1)/4$
$T(d)$	$d(d^2-1)/12$
$C_{LLS_1S_2}^{ij;AB}$	$L_i L_j S_1^A S_2^B$
C_{ijab}	$L_i L_j \phi_a \phi_b$
C_5^{ij}	$(L_i^T C L_j) H H$
R_W	$16\pi^2 \beta_{C_5} \supset R_W \lambda_5 C_{LLS_1S_2}$

General-interaction coupling index

Coupling	Field content / definition	Contraction / source

Each X_i labels one field-content class in the general-quartic registry; a superscript distinguishes different invariant contractions where present. The original RGBeta identifier and contraction are retained in the index. This is notation only, not a physical normalisation mapping.

Complete BSM UV Lagrangian

Matchete BSM Lagrangian, excluding the pure Standard Model contribution. $\mathcal{L}_{\text{BSM}} = \text{Coupling}(\text{MS1}, \{\}, 0)^2\text{Field}(\text{NewScalar1},$

+Coupling(lambdaH1Inv1, {}, 0)Field(NewScalar1, Scalar, {}, {})Bar(Field(NewScalar1, Scalar, {}, {}))Field(H, Scalar, {Index(d$$2, SU2L(fund))}, {})CG(T3HiggsPortal1CGInv1, {Index(d$$1, SU2L(fund)), Bar(Index(d$$2, SU2L(fund)))})Bar(Field(H, Scalar, {Index(d$$1, SU2L(fund))}, {}))

+Coupling(lambdaT3, {}, 0)Field(NewScalar1, Scalar, {}, {})Bar(Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {}))Field(H, Scalar, {Index(d$$1, SU2L(fund))}, {})Field(H, Scalar, {Index(d$$2, SU2L(fund))}, {})CG(T3MixCG, {Bar(Index(d$$1, SU2L(fund))), Bar(Index(d$$2, SU2L(fund))), Index(d$$1, T3BSMd3)})

$$+\frac{1}{2}\text{Coupling}(\text{lambdaS1Inv1}, \{\}, 0)\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\}, \{\})^2\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\}, \{\}))^2$$

$$+\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\}, \{\text{Index}(\text{d\$\$1}, \text{Lorentz})\})\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\}, \{\text{Index}(\text{d\$\$1}, \text{Lorentz})\}))$$

+Bar(Coupling(lambdaT3, {}, 0))Bar(Field(NewScalar1, Scalar, {}, {}))Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {})Bar(Field(H, Scalar, {Index(d$$1, SU2L(fund))}, {}))Bar(Field(H, Scalar, {Index(d$$2, SU2L(fund))}, {}))CG(Bar(T3MixCG), {Index(d$$1, SU2L(fund)), Index(d$$2, SU2L(fund)), Index(d$$1, T3BSMd3)})

$$+\text{Coupling}(\text{MS2}, \{\}, 0)^2\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\})(-\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\})))$$

$$+\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\text{Index}(\text{d\$\$1}, \text{Lorentz})\})\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\text{Index}(\text{d\$\$1}, \text{Lorentz})\}))$$

+$\frac{1}{2}$Coupling(lambdaH2Inv1, {}, 0)Field(NewScalar2, Scalar, {Index(d$$2, T3BSMd3)}, {})Bar(Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {}))Field(H, Scalar, {Index(d$$2, SU2L(fund))}, {})Bar(Field(H, Scalar, {Index(d$$1, SU2L(fund))}, {}))CG(T3HiggsPortal2CGInv1, {Index(d$$1, SU2L(fund)), Bar(Index(d$$2, SU2L(fund)))}, Index(d$$1, T3BSMd3), Index(d$$2, T3BSMd3))

+$\frac{1}{2}$Coupling(lambdaH2Inv1, {}, 0)Field(NewScalar2, Scalar, {Index(d$$2, T3BSMd3)}, {})Bar(Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {}))Field(H, Scalar, {Index(d$$2, SU2L(fund))}, {})Bar(Field(H, Scalar, {Index(d$$1, SU2L(fund))}, {}))CG(T3HiggsPortal2CGInv1, {Index(d$$1, SU2L(fund)), Bar(Index(d$$2, SU2L(fund)))}, Index(d$$2, T3BSMd3), Index(d$$1, T3BSMd3))

+$\frac{1}{2}$Coupling(lambdaH2Inv2, {}, 0)Field(NewScalar2, Scalar, {Index(d$$2, T3BSMd3)}, {})Bar(Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {}))Field(H, Scalar, {Index(d$$2, SU2L(fund))}, {})Bar(Field(H, Scalar, {Index(d$$1, SU2L(fund))}, {}))CG(T3HiggsPortal2CGInv2, {Index(d$$1, SU2L(fund)), Bar(Index(d$$2, SU2L(fund)))}, Index(d$$1, T3BSMd3), Index(d$$2, T3BSMd3))

+$\frac{1}{2}$Coupling(lambdaH2Inv2, {}, 0)Field(NewScalar2, Scalar, {Index(d$$2, T3BSMd3)}, {})Bar(Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {}))Field(H, Scalar, {Index(d$$2, SU2L(fund))}, {})Bar(Field(H, Scalar, {Index(d$$1, SU2L(fund))}, {}))CG(T3HiggsPortal2CGInv2, {Index(d$$1, SU2L(fund)), Bar(Index(d$$2, SU2L(fund)))}, Index(d$$2, T3BSMd3), Index(d$$1, T3BSMd3))

+Coupling(lambda12Inv1, {}, 0)Field(NewScalar1, Scalar, {}, {})Bar(Field(NewScalar1, Scalar, {}, {}))CG(T3CrossScalarCGInv1, {Index(d$$1, T3BSMd3), Index(d$$2, T3BSMd3)})Field(NewScalar2, Scalar, {Index(d$$2, T3BSMd3)}, {})Bar(Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {}))

+$\frac{1}{2}$Coupling(lambdaS2Inv1, {}, 0)Field(NewScalar2, Scalar, {Index(d$$3, T3BSMd3)}, {})Field(NewScalar2, Scalar, {Index(d$$4, T3BSMd3)}, {})Bar(Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {}))Bar(Field(NewScalar2, Scalar, {Index(d$$2, T3BSMd3)}, {}))CG(T3Scalar2SelfCGInv1, {Index(d$$1, T3BSMd3), Index(d$$3, T3BSMd3)}, Index(d$$2, T3BSMd3), Index(d$$4, T3BSMd3))

+$\frac{1}{2}$Coupling(lambdaS2Inv2, {}, 0)Field(NewScalar2, Scalar, {Index(d$$3, T3BSMd3)}, {})Field(NewScalar2, Scalar, {Index(d$$4, T3BSMd3)}, {})Bar(Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {}))Bar(Field(NewScalar2, Scalar, {Index(d$$2, T3BSMd3)}, {}))CG(T3Scalar2SelfCGInv2, {Index(d$$1, T3BSMd3), Index(d$$3, T3BSMd3)}, Index(d$$2, T3BSMd3), Index(d$$4, T3BSMd3))

−Coupling(MF, {}, 0)NCM(Bar(Field(NewFermion, Fermion, {Index(d$$1, T3BSMd2), Index(d$$1, NFlavor)}, {})), Field(NewFermion, Fermion, {Index(d$$1, T3BSMd2), Index(d$$1, NFlavor)}, {}))

+Coupling(y2, {Index(d$$1, Flavor), Index(d$$1, NFlavor)}, 0)Bar(Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {}))CG(T3Y2CG, {Index(d$$1, SU2L(fund)), Bar(Index(d$$1, T3BSMd2)), Index(d$$1, T3BSMd3)})NCM(Bar(Field(I, Fermion, {Index(d$$1, SU2L(fund)), Index(d$$1, Flavor)}, {})), DiracProduct(Proj(1)), Field(NewFermion, Fermion, {Index(d$$1, T3BSMd2), Index(d$$1, NFlavor)}, {}))

+Field(NewScalar1, Scalar, {}, {})Coupling(y1, {Index(d$$1, Flavor), Index(d$$1, NFlavor)}, 0)CG(T3Y1CG, {Index(d$$1, SU2L(fund)), Index(d$$1, T3BSMd2)})NCM(Bar(Field(I, Fermion, {Index(d$$1, SU2L(fund)), Index(d$$1, Flavor)}, {})), DiracProduct(GammaCC, Proj(1)), Transp(Bar(Field(NewFermion, Fermion, {Index(d$$1, T3BSMd2), Index(d$$1, NFlavor)}, {}))))

+iNCM(Bar(Field(NewFermion, Fermion, {Index(d$$1, T3BSMd2), Index(d$$1, NFlavor)}, {})), DiracProduct(GammaM(Index(d$$1, Lorentz))), Field(NewFermion, Fermion, {Index(d$$1, T3BSMd2), Index(d$$1, NFlavor)}, {Index(d$$1, Lorentz)}))

+Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {})Bar(Coupling(y2, {Index(d$$1, Flavor), Index(d$$1, NFlavor)}, 0)CG(Bar(T3Y2CG), {Bar(Index(d$$1, SU2L(fund))), Index(d$$1, T3BSMd2), Index(d$$1, T3BSMd3)})NCM(Bar(Field(NewFermion, Fermion, {Index(d$$1, T3BSMd2), Index(d$$1, NFlavor)}, {})), DiracProduct(Proj(−1)), Field(I, Fermion, {Index(d$$1, SU2L(fund)), Index(d$$1, Flavor)}, {}))

+Bar(Field(NewScalar1, Scalar, {}, {}))Bar(Coupling(y1, {Index(d$$1, Flavor), Index(d$$1, NFlavor)}, 0)CG(Bar(T3Y1CG), {Bar(Index(d$$1, SU2L(fund))), Bar(Index(d$$1, T3BSMd2))})NCM(Transp(Field(I, Fermion, {Index(d$$1, SU2L(fund)), Index(d$$1, Flavor)}, {})), DiracProduct(GammaCC, Proj(−1)), Field(NewFermion, Fermion, {Index(d$$1, T3BSMd2), Index(d$$1, NFlavor)}, {}))

Interaction inventory

- `Yukawa1_GenericCG`
- `Yukawa2_GenericCG`
- `Scalar1SelfInv1`
- `Scalar2SelfInv1`
- `Scalar2SelfInv2`
- `HiggsPortal1Inv1`
- `HiggsPortal2Inv1`
- `HiggsPortal2Inv2`
- `ScalarCrossPortalInv1`
- `T3ScalarMix_GenericCG`

One-loop renormalisable beta functions

$16\pi^2\, dX/d\ln\mu = \beta_X$

g_1

$$16\pi^2 \frac{dg_1}{d\ln\mu} = \frac{163{g_1}^3}{6}$$

g_2

$$16\pi^2 \frac{dg_2}{d\ln\mu} = -\frac{{g_2}^3}{2}$$

g_3

$$16\pi^2 \frac{dg_3}{d\ln\mu} = -7{g_3}^3$$

Y_u

$$\begin{aligned}
16\pi^2 \frac{dY_u}{d\ln\mu} = {} & 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_u)_{ij} \\
& + \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_u)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_u)_{ij} \\
& - \frac{3}{2}(\Big(Y_d\ (Y_d)^\dagger\ Y_u\Big)_{ij} - \Big(Y_u\ (Y_u)^\dagger\ Y_u\Big)_{ij}) \\
& - \frac{9}{4}{g_2}^2\,(Y_u)_{ij} \\
& - 8{g_3}^2\,(Y_u)_{ij} \\
& - \frac{17}{12}{g_1}^2\,(Y_u)_{ij}
\end{aligned}$$

Y_d

$$\begin{aligned}
16\pi^2 \frac{dY_d}{d\ln\mu} = {} & \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij} \\
& + \frac{3}{2}(\Big(Y_d\ (Y_d)^\dagger\ Y_d\Big)_{ij} - \Big(Y_u\ (Y_u)^\dagger\ Y_d\Big)_{ij}) \\
& - \frac{9}{4}{g_2}^2\,(Y_d)_{ij} \\
& - 8{g_3}^2\,(Y_d)_{ij} \\
& - \frac{5}{12}{g_1}^2\,(Y_d)_{ij}
\end{aligned}$$

Y_e

$$\begin{aligned}16\pi^2 \frac{dY_e}{d\ln\mu} &= 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_e)_{ij} \\ &+ 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_e)_{ij} \\ &+ \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_e)_{ij} \\ &+ \frac{1}{4}(2\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij} + 3\left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij} + 6\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij}) \\ &- \frac{9}{4}{g_2}^2\,(Y_e)_{ij} \\ &- \frac{15}{4}{g_1}^2\,(Y_e)_{ij}\end{aligned}$$

y_1

$$\begin{aligned}16\pi^2 \frac{dy_1}{d\ln\mu} &= 2T_\nu^{(1)}\,(y_1)_{ij} \\ &+ \frac{3}{4}\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij} \\ &+ \frac{1}{2}\left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij} \\ &+ \left(y_1\ (y_1)^\dagger\ y_1\right)_{ij} \\ &- \frac{9}{2}{g_2}^2\,(y_1)_{ij} \\ &- \frac{15}{2}{g_1}^2\,(y_1)_{ij}\end{aligned}$$

y_2

$$\begin{aligned}16\pi^2 \frac{dy_2}{d\ln\mu} &= T_\nu^{(2)}\,(y_2)_{ij} \\ &+ \frac{1}{2}(\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij} + \left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij} + 3\left(y_2\ (y_2)^\dagger\ y_2\right)_{ij}) \\ &- \frac{9}{2}{g_2}^2\,(y_2)_{ij} \\ &- \frac{15}{2}{g_1}^2\,(y_2)_{ij}\end{aligned}$$

M_F

$$\begin{aligned}16\pi^2 \frac{dM_F}{d\ln\mu} &= \frac{1}{4}(2\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij} + 3\left(M_F\ (y_2)^\dagger\ y_2\right)_{ij}) \\ &- \frac{9}{2}{g_2}^2\,(M_F)_{ij} \\ &- \frac{27}{2}{g_1}^2\,(M_F)_{ij}\end{aligned}$$

m_1^2

$$16\pi^2 \frac{dm_1^2}{d\ln\mu} = m_1^2(4T_\nu^{(1)} + 4\lambda_{S_1}^{(1)})$$

$$
\begin{aligned}
&- 8\,\mathrm{Tr}\Big(M_F\ (M_F)^\dagger\ (y_1)^T\ (y_1)^*\Big)\\
&- 24{g_1}^2 m_1^2\\
&+ 6\lambda_{12}^{(1)} m_2^2
\end{aligned}
$$

m_2^2

$$
\begin{aligned}
16\pi^2\,\frac{dm_2^2}{d\ln\mu} = m_2^2(2T_\nu^{(2)} + 8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})\\
&- 4\,\mathrm{Tr}\Big(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\Big)\\
&- 12{g_2}^2 m_2^2\\
&- 6{g_1}^2 m_2^2\\
&+ 2\lambda_{12}^{(1)} m_1^2
\end{aligned}
$$

λ_1

$$
\begin{aligned}
16\pi^2\,\frac{d\lambda_1}{d\ln\mu} = \lambda_1(12\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 4\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 12\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big))\\
&- 12\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\Big)\\
&- 4\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\Big)\\
&- 12\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\Big)\\
&+ 4\lambda_5\,(\lambda_5)^*\\
&+ \frac{9{g_2}^4}{4}\\
&+ {g_1}^2\left(\frac{3{g_2}^2}{2} - 3\lambda_1\right)\\
&- 9{g_2}^2\lambda_1\\
&+ \frac{3{g_1}^4}{4}\\
&+ 12{\lambda_1}^2\\
&+ 2{\lambda_{HS_1}^{(1)}}^2\\
&+ 6{\lambda_{HS_2}^{(1)}}^2\\
&+ 3{\lambda_{HS_2}^{(A)}}^2
\end{aligned}
$$

$\lambda_{S_1}^{(1)}$

$$
\begin{aligned}
16\pi^2\,\frac{d\lambda_{S_1}^{(1)}}{d\ln\mu} = 8\lambda_{S_1}^{(1)} T_\nu^{(1)}\\
&- 8\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\Big)\\
&+ 192{g_1}^4\\
&- 48{g_1}^2\lambda_{S_1}^{(1)}\\
&+ 6{\lambda_{12}^{(1)}}^2\\
&+ 4{\lambda_{HS_1}^{(1)}}^2
\end{aligned}
$$

$$+ 10{\lambda_{S_1}^{(1)}}^2$$

$\lambda_{S_2}^{(1)}$

$$\begin{aligned}
16\pi^2 \frac{d\lambda_{S_2}^{(1)}}{d\ln\mu} = {} & \lambda_{S_2}^{(1)}(4T_\nu^{(2)} + 8\lambda_{S_2}^{(A)}) \\
& - \frac{8}{3}\operatorname{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big) \\
& + 22{g_2}^4 \\
& + {g_1}^2\left(8{g_2}^2 - 12\lambda_{S_2}^{(1)}\right) \\
& - 24{g_2}^2\lambda_{S_2}^{(1)} \\
& + 12{g_1}^4 \\
& + 2{\lambda_{12}^{(1)}}^2 \\
& + 4{\lambda_{HS_2}^{(1)}}^2 \\
& + 14{\lambda_{S_2}^{(1)}}^2 \\
& + 16{\lambda_{S_2}^{(A)}}^2
\end{aligned}$$

$\lambda_{HS_1}^{(1)}$

$$\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_1}^{(1)}}{d\ln\mu} = {} & \lambda_{HS_1}^{(1)}(4T_\nu^{(1)} + 6\operatorname{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\operatorname{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\operatorname{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)) \\
& - 4\operatorname{Tr}\Big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\Big) \\
& + 12\lambda_5\,(\lambda_5)^* \\
& - \frac{9{g_2}^2\lambda_{HS_1}^{(1)}}{2} \\
& + 12{g_1}^4 \\
& - \frac{51{g_1}^2\lambda_{HS_1}^{(1)}}{2} \\
& + 6\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)} \\
& + 6\lambda_1\lambda_{HS_1}^{(1)} \\
& + 4{\lambda_{HS_1}^{(1)}}^2 \\
& + 4\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}
\end{aligned}$$

$\lambda_{HS_2}^{(1)}$

$$\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_2}^{(1)}}{d\ln\mu} = {} & \lambda_{HS_2}^{(1)}(2T_\nu^{(2)} + 6\operatorname{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\operatorname{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\operatorname{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) + 4\lambda_{S_2}^{(A)}) \\
& - 2\operatorname{Tr}\Big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\Big) \\
& + 4\lambda_5\,(\lambda_5)^* \\
& + 6{g_2}^4 \\
& - \frac{33{g_2}^2\lambda_{HS_2}^{(1)}}{2}
\end{aligned}$$

$$
\begin{aligned}
&+ 3{g_1}^4 \\
&- \frac{15{g_1}^2\lambda_{HS_2}^{(1)}}{2} \\
&+ 2\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)} \\
&+ 6\lambda_1\lambda_{HS_2}^{(1)} \\
&+ 4{\lambda_{HS_2}^{(1)}}^2 \\
&+ 8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)} \\
&+ 6{\lambda_{HS_2}^{(A)}}^2
\end{aligned}
$$

$\lambda_{12}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{12}^{(1)}}{d\ln\mu} = \lambda_{12}^{(1)}(&4T_\nu^{(1)} + 2T_\nu^{(2)} + 4\lambda_{S_2}^{(A)}) \\
&- 4\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big) \\
&+ 4\lambda_5\,(\lambda_5)^* \\
&- 12{g_2}^2\lambda_{12}^{(1)} \\
&+ 48{g_1}^4 \\
&- 30{g_1}^2\lambda_{12}^{(1)} \\
&+ 4{\lambda_{12}^{(1)}}^2 \\
&+ 4\lambda_{12}^{(1)}\lambda_{S_1}^{(1)} \\
&+ 8\lambda_{12}^{(1)}\lambda_{S_2}^{(1)} \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}
\end{aligned}
$$

λ_5

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_5}{d\ln\mu} = \lambda_5(&2T_\nu^{(1)} + T_\nu^{(2)} + 6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) + 4\lambda_{HS_2}^{(A)}) \\
&- \frac{21{g_2}^2\lambda_5}{2} \\
&- \frac{33{g_1}^2\lambda_5}{2} \\
&+ 2\lambda_{12}^{(1)}\lambda_5 \\
&+ 2\lambda_1\lambda_5 \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_5 \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_5
\end{aligned}
$$

$\lambda_{HS_2}^{(A)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_2}^{(A)}}{d\ln\mu} = \lambda_{HS_2}^{(A)}(&2T_\nu^{(2)} + 6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) + 16\lambda_{S_2}^{(A)}) \\
&+ \frac{4}{3}\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\Big) \\
&+ \frac{8}{3}\lambda_5\,(\lambda_5)^*
\end{aligned}
$$

$$
\begin{aligned}
&+ {g_1}^2 \left(-4{g_2}^2 - \frac{15\lambda_{HS_2}^{(A)}}{2} \right) \\
&- \frac{33{g_2}^2\lambda_{HS_2}^{(A)}}{2} \\
&+ 2\lambda_1\lambda_{HS_2}^{(A)} \\
&+ 8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)} \\
&+ 2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}
\end{aligned}
$$

$\lambda_{S_2}^{(A)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{S_2}^{(A)}}{d\ln\mu} = {} & 4\lambda_{S_2}^{(A)} T_\nu^{(2)} \\
& - \frac{2}{3}\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger \Big) \\
& - 2{g_2}^4 \\
& + {g_1}^2 \left(8{g_2}^2 - 12\lambda_{S_2}^{(A)} \right) \\
& - 24{g_2}^2\lambda_{S_2}^{(A)} \\
& + 3{\lambda_{HS_2}^{(A)}}^2 \\
& + 12\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)} \\
& + 20{\lambda_{S_2}^{(A)}}^2
\end{aligned}
$$